Sum and Difference Formulas

Exact values with sum and difference formulas

1 Use a sum or difference formula to find the exact value of $\cos 75°$.

2 Find the exact value of $\sin \frac{\pi}{12}$.

3 Find the exact value of $\tan 105°$.

Working with given values

4 Angles A and B are both in Quadrant I with $\sin A = \frac{4}{5}$ and $\cos B = \frac{5}{13}$. Find:

a $\sin(A + B)$

b $\cos(A + B)$

c the quadrant of $A + B$

5 Use a sum or difference formula to simplify:

a $\sin\left(x + \frac{\pi}{2}\right)$

b $\cos(x - \pi)$

6 Angles A and B are both in Quadrant II with $\sin A = \frac{8}{17}$ and $\sin B = \frac{3}{5}$. Find $\cos(A - B)$ exactly.

Verifying identities using sum and difference formulas

7 Verify the identity $\cos(x + y) + \cos(x - y) = 2\cos x \cos y$.

8 Verify the identity $\sin(x + y) - \sin(x - y) = 2\cos x \sin y$.

9 Show that $\tan\left(x + \frac{\pi}{4}\right) = \frac{1 + \tan x}{1 - \tan x}$ using the tangent sum formula $\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$.

Applications

10 Two musical tones combine to form a sound wave modeled by $y = \sin\left(x + \frac{\pi}{6}\right)$. Expand this expression using the sum formula, and evaluate the coefficient of $\cos x$ exactly.

Additional practice

11 Find the exact value of $\cos\frac{\pi}{12}$.

12 Angles A and B satisfy $\tan A = \frac{3}{4}$ (QI) and $\tan B = \frac{5}{12}$ (QI). Find $\tan(A + B)$ using the tangent sum formula.

13 Simplify $\cos\left(\frac{\pi}{2} - x\right)\cos x + \sin\left(\frac{\pi}{2} - x\right)\sin x$ using the cosine difference formula in reverse.